FCM-GA-Based Vessel Segmentation in Retinal Imaging

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Abstract. The distribution of blood vessels in the retinal image is important in diagnosing and follow-up of many eye diseases and disorders, such as diabetic retinopathy, glaucoma, and hypertension. Traditional segmentation algorithms often have difficulty dealing with the complexity of retinal images, especially when there is noise, irregularity, and unstable arterial thickness. This paper presents a method that combines FCM (fuzzy C-means clustering) with genetic algorithm (GA) improvement to enhance the accuracy and consistency of vessel segmentation. FCM can control the uncertainty and variability of retinal images by grouping pixels into different groups, thus making the vessels more accurate. Then, GA improves face segmentation by increasing the dynamic function based on Dice's similarity coefficient, which results in higher boundary accuracy and fewer false positives. The results show that the proposed method outperforms FCM alone, as shown by higher scores and Intersection of Unity (IoU) on retinal images.

INTRODUCTION

Identifying and monitoring a variety of ocular and systemic disorders requires segmenting blood vessels from retinal imaging because it allows for identifying and quantifying retinal vascular structures, which is required for detecting disorders such as diabetic retinopathy, glaucoma, and hypertension. Accurate segmenting of these veins is critical for effective treatment planning and monitoring disease progression. Image segmentation, which divides a digital image into a set of disjoint regions, has numerous applications in computer vision (segmenting and analysing facial features, fingerprints, and so on), medical imaging (MRI, CT, ultrasound), tumour identification, segmenting blood vessels in angiographic images, and so on. [1-3]. Several conventional segmentation methods exist, such as thresholding [4], edge detection, watersheds [5], histogram-based, region-based [6], clustering-based segmentation [7], and others. Although several conventional techniques for vessel segmentation exist, they frequently struggle with the complexity and diversity of retinal pictures, particularly when dealing with images with noise, poor contrast, varying vessel thickness and the ability to change contrast environment. Consequently, there has been significant interest in leveraging advanced computational techniques for enhancing segmentation accuracy. A promising approach for vascular segmentation is the combination of Genetic Algorithms (GA) and Fuzzy C-Means (FCM) clustering. Genetic algorithms are modified heuristic search algorithms based on natural selection and genetics. They provide a solid foundation for solving optimization problems. GA is useful for exploring large search areas and improving segmentation solutions by first re-optimizing the set of candidate solutions using the power function [8-10]. Fuzzy C-means clustering, on the other hand, is an effective image segmentation technique that assigns individual data points to groups based on their group membership rather than a strict binary distribution. This merging method is particularly suitable for handling blur and noise in retinal images because it allows some of the members to be in different groups, allowing for more detail and detailed broad segmentation. Improved retinal vessel segmentation using average clustering with genetic algorithms. Genetic algorithm-based approach to reconstruct the population of segmentation solutions to increase the power of prioritization. This exercise is designed to increase the balance between similar segments and dissimilar segments, improving the accuracy and clarity of the vessels.

LITERATURE REVIEW

Medical image segmentation aims to enhance the clarity of changes in anatomical or pathogenic structures in images; the significant gains in diagnostic precision and efficacy often serve an integral role in computer-aided diagnosis and smart medicine [11]. Typical segmentation tasks involve analysing X-rays, CT scans, and mammography images, diagnosing malignant polyps, ulcers, IBS, and other disorders, as well as segmenting livers [12, 13], brain tumours [14, 15], optical discs [16, 17], and more. To enhance the accuracy and help the clinicians to make an accurate decision for the analysis, it is necessary to divide the key components of the image and provide the main information of the material. Conventional methods for segmentation involve edge detection, clustering, and region-based segmentation and so on. In [18], a hybrid method is proposed for accurate segmentation of ultrasound medical images, which exploits the features of kernel fuzzy clustering with spatial constraints (KFCM-S) and edge-based edge clustering using distance as contour function of regular layer (DRL). Results are obtained between synthetic images and real ultrasound images. Proposed method-1 has achieved an accuracy of 92.91% and proposed method-2 has achieved 93.15% on all ultrasound images. To acquire reliable segmentation, edge information is often ignored in favor of primary region recognition in most approaches. To achieve precise segmentation, [19] introduced the (ET-NET) Edge-a-tension guidance network, a general medical segmentation technique that embeds edge-attention representation to a direct segmentation network. To discuss the challenges of image segmentation in computer vision, particularly its inability to adapt to real-world changes in images [20] introduced a closed-loop image segmentation that employs a genetic algorithm to optimize segmentation parameters in response to varying environmental conditions. The research focuses on the timing of outdoor photographs, including changes in light (sun) position, light intensity, and weather conditions. The results show that transform image segmentation can produce good results (95%) with a minimum number of segmentation steps. The solution consistently improves performance by more than 30% compared to traditional methods or preset segmentation parameters. [21] A new method is proposed that includes Segmenting FCM and using hybrid GA and PSO. MRI image segmentation is accomplished using the suggested technique. This method's output is contrasted with that of basic segmentation techniques like FCM and KFCM by utilizing parameters like transformation, global error, and Rand index. Experiments verify that this approach is more successful and efficient. A technique for brain imaging segmentation was proposed by combining FCM with a rough set [22]. The suggested method improves image segmentation by creating an attribute value table from Fuzzy C-Means (FCM) results with different cluster numbers, and then merging regions based on calculated attribute weights and similarity evaluations. The method was validated on simulated images, brain CT and MRI and outperformed FCM alone, especially in segmentation in non-visible regions. Using FCM, the author proposed HisFCM in [23-28], an improved FCM algorithm that uses image histograms and works in two stages: first, select the time to calculate the second group centroid. It is used instead of FCM for segmentation. Compared with FCM and other variants, HisFCM provides better performance while maintaining the accuracy of the comparison. Clinical experiments show that He-FCM can achieve good classification in less than 0.1 seconds and meets the processing time of medical images.

METHODOLOGY

This section provides an overview of genetic algorithm (GA) techniques for vessel classification in retinal images. The process involves a series of procedural steps from data processing to obtaining the most suitable segmentation using genetic algorithms. The main stages are loading and preparing the retinal image, starting the genetic algorithm, evaluating the suitability of the segmentation solutions, selecting and generating new solutions by rotation and transformation, and then using the most suitable solution for vessel segmentation. In the following chapter, each step will be explained in detail so that the reader can understand the method. Figure 1 shows the flowchart of the retinal vessel segmentation method. It presents a sequential process starting with the preprocessing of the retinal image and ending with the processing of the genetic algorithm through fuzzy C-means clustering.

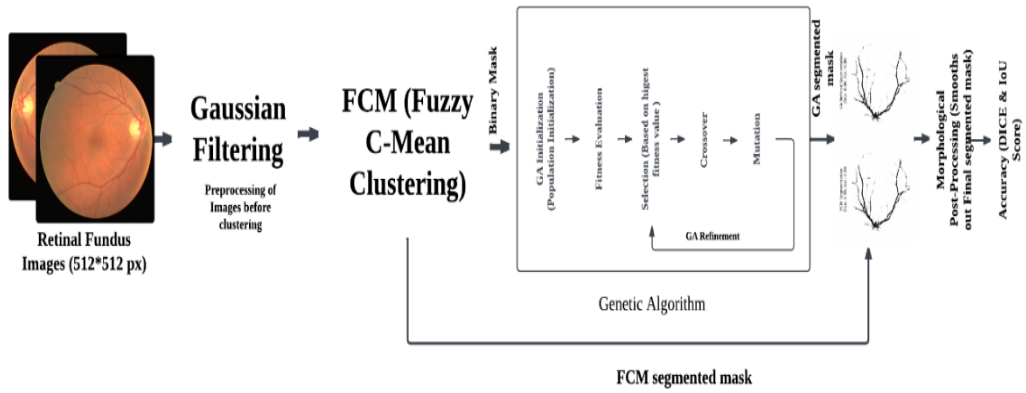


FIGURE 1. Flowchart illustrating the methodology for retinal vessel segmentation using Fuzzy C-Means (FCM) clustering and a Genetic Algorithm (GA) [Source: Author].

DATASET

Loading retinal images and corresponding masks is the first step in our process of vessel segmentation from retinal images. The segmentation process depends on these images because they provide the necessary information for training and evaluation. Vessel masking is important for the segmentation process because it provides several important factors that can improve and guide the chromosome segmentation of the affected regions. This file contains 80 high-resolution retinal fundus images captured using state-of-the-art technology. Each image has a resolution of 512x512 pixels and comes with pixel-level ground truth annotations to determine the location of vessels. These annotations are important for designing and evaluating decision segmentation algorithms because they provide useful context for comparing and validating segmentation results.

GAUSSIAN FILTERING

Gaussian noise is a common problem in retinal imaging, especially fundus imaging, resulting from sensor imperfecti ons, poor illumination, and electrical interference. These noises can distort reality by revealing important details suc h as small blood vessels and subtle signs of pathology. Gaussian filtering effectively reduces this noise, improving i mage quality, and thereby improving visualization of important anatomical structures.

$$G(x,\ y)\ =\ \frac{1}{2\pi\sigma^2}\ \ exp\, exp\left(-\frac{x^2\ +\ y^2}{2\sigma^2}\right)$$

The filtered image $I_{fltered}(x,y)$ is obtained by convolving the original image $I(x,y)$ with a Gaussian kernel:

$$I_{\text{filtered}}(x,\ y) = I(x,\ y) * G(x,\ y)$$

FIGURE 2. Comparison of retinal images filtered with Gaussian kernels of varying sigma values [Source: Author].

Increasing the sigma value in a Gaussian filter improves blurring and noise reduction, however, it will also cause the loss of fine points and edge points as shown in Figure 2. Higher sigma values produce a smoother, softer appearance for the image.

FCM - FUZZY C-MEAN CLUSTERING

After filtering and normalizing the image, use FCM grouping to divide the pre-processed image into different groups based on pixel values. Sigma value '1' was chosen because it provides the best compromise between effective noise reduction and keeping crucial image details, ensuring that essential structures stay recognizable while minimizing high-frequency noise. The FCM algorithm minimizes the objective function.

$$J(U, C) = \sum_{i=1}^{N} \sum_{j=1}^{c} u_{ij}^{m} |x_i - c_j|^2$$

- u_{ij}: membership degree of pixel x_i to cluster j,
- c_j: centroid of cluster j,
- m: fuzziness parameter (typically $m > 1$).

The membership values u_{ij} are iteratively calculated and the pixel intensities c are clustered into groups. Obtain a membership map U_j for each cluster, where each map represents the membership of each pixel to the corresponding cluster:

$$U_j(x, y) = u_{ij}$$

GENETIC ALGORITHM

Our vascular segmentation approach relies primarily on a genetic algorithm (GA) to maximise retinal image segmentation. GAare modified heuristic search algorithms based on generative techniques such as natural selection and genetics. To maximize a specific fitness function, the GA iteratively refines a population of possible segmentation solutions. After applying Fuzzy C-Means (FCM) clustering to the preprocessed retinal images, the pixel intensities are clustered into c distinct clusters. Each cluster is associated with a membership map $U_j(x, y)$, where each pixel has a degree of membership to a specific cluster. The membership values u_{ij} are computed iteratively, ensuring that pixels belonging to similar regions (such as blood vessels, background, or other anatomical structures) are grouped. Following the FCM clustering, the segmentation process proceeds with the initialization of the genetic algorithm:

POPULATION INITIALIZATION

The initial population of the GA consists of binary partitioning masks derived from the membership graph generated by the FCM. Each individual in the population represents a segmentation solution and corresponds to a group. Mathematically, the initial population P can be defined as:

$$P = \{I_1, I_2. . . I_N\}$$

Where; I_n represents an individual binary mask in the population, and N represents population size.

FITNESS EVALUATION

GA uses a power function based on Dice's correlation coefficient to evaluate the confidence of each individual in the population. This coefficient measures the overlap between the binary segmentation cover and the ground truth cover G and is expressed as:

$$Dice(I_n, G) = \frac{2 \times |I_n \cap G|}{|I_n| + |G|}$$

Where $|I_n \cap G|$ denotes the number of pixels at the intersection of I_n and G, and $|I_n|$ and $|G|$ denotes the number of pixels in I_n and G, respectively. The goal of the GA is to maximize this Dice coefficient, thereby refining the segmentation mask to closely match the ground truth.

SELECTION, CROSSOVER and MUTATION

Selection: Individuals with higher physical fitness will be selected for reproduction. The selection process can be modelled using competitive selection methods, which select the best individuals from a subset of the population.
Crossover: Selected individuals are crossed over to produce offspring. This process combines the features of two-parent masks to create a new mask. The crossover operation can be represented as:

$$\text{Offspring} = \text{Crossover}\ (I_{n1},\ I_{n2})$$

Where; I_{n1} and I_{n2} are parent masks.

Mutation: To maintain diversity within the population, a mutation operation is applied, which randomly flips bits in the binary mask with a certain probability. This mutation can be expressed as:

$I'_n = \text{Mutate}\ (I_n)$

Where; I'_n is the mutated version of the individual I_n.

GA REFINEMENT AND FINAL SEGMENTATION

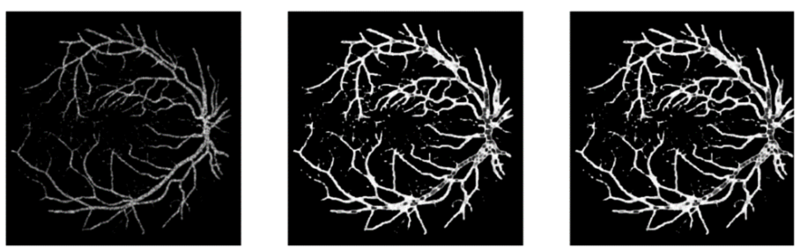
Improvement in genetic algorithms requires the population to undergo selection, competition, and mutation over generations, ultimately transforming the segmentation mask into the best solution. After many generations, the genetic algorithm converges to the best individuals in the population, which represents the final segmented image. The final segmentation mask should have a high Dice coefficient that matches the ground truth well. This involves the use of morphological functions (such as the closed morphological function) to eliminate small gaps and smooth boundaries:

$$S_{\text{processed}} = \text{MorphClose}(S)$$

Where; S is the segmentation mask, and MorphClose is a morphological closing operation.

RESULTS ANALYSIS

The segmentation process first converts the retinal image to a gray image and then performs Gaussian filtering to reduce noise and improve the main structure. Then, the pixel values are smoothed and FCM clustering is used to create membership maps representing different image regions. These maps are the initial population for a genetic algorithm (GA) that iteratively performs segmentation by increasing the quality score based on metrics such as dice coefficients. Finally, post-processing techniques such as morphological operations are used to enhance the segmented image to identify the vessels. Figure 3 represents the original, FCM and GA refined segmented image.



(a) (b) (c)

FIGURE 3. Retinal Vessel Image Segmentation. (a) represents an original image, (b) represents a segmented image using Fuzzy C-mean Clustering (FCM), and (c) represents a segmented image after GA refinement [Source: Author].

After the segmentation process, the retinal image segmentation method is evaluated using the dice coefficient and Intersection over Union (IoU), two commonly used metrics in image segmentation software.

Dice coefficient

The dice coefficient is a measure of the overlap between the estimated segmentations and the ground truth. The formula is:

$$Dice\ Coefficient = \frac{2 \times |AintersectionB|}{|A| + |B|}$$

Here, A represents the pixel group in the segmented image and B represents the pixel group in the ground truth. The dice coefficient varies from 0 to 1, where 1 represents perfect overlap. However, after using genetic algorithm (GA) optimization, the dice coefficients of all images were significantly improved. The genetic algorithm effectively improves the accuracy of the boundary and reduces the negative, thus approaching the ground truth and demonstrating the optimization capabilities.

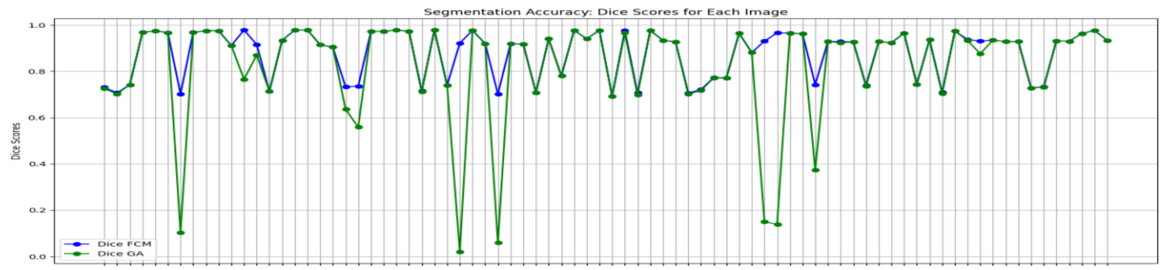


FIGURE 4. DICE Score for segmented images using FCM-GA [Source: Author].

Intersection over Union (IoU)

The Intersection over Union (IoU) metric provides another measure of segmentation accuracy, defined as:

$$IoU = \frac{|AIntersectionB|}{|AUnionB|}$$

Where; A is the set of pixels in the segmented image and B is the set of pixels in the ground truth. Like the dice coefficient, IoU varies between 0 and 1, with higher values indicating better segmentation. However, with the GA improvement, the IoU scores of all images are significantly improved. The genetic algorithm effectively balances the real vessel pixels while excluding non-vessel pixels, resulting in better overlap and fewer irrelevant pixels in the segmented region.

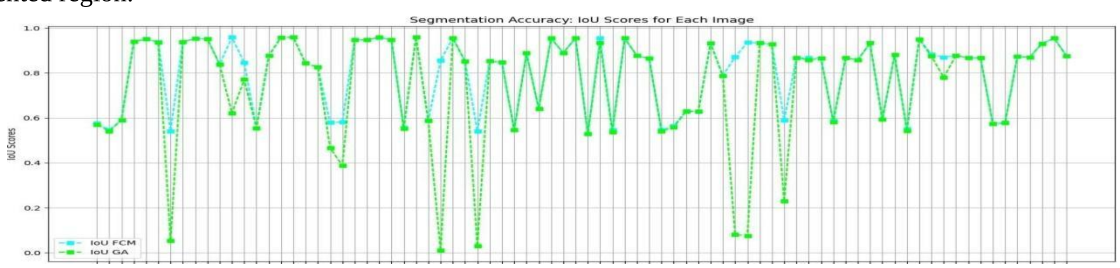


FIGURE 5. IoU Score for segmented images using FCM-GA [Source: Author].

The combination of Fuzzy C-means (FCM) clustering and genetic algorithm (GA) refinement similarly improves the image segmentation performance, as shown by the improvements in Dice and Cross-Union (IoU) scores in Figures 4 and 5. GA refinement not only improves the accuracy but also improves the segmentation results across different images. In difficult cases where using FCM alone may be problematic, GA provides accurate and reliable segmentation by improving Dice and IoU scores. This method becomes a good way to improve segmentation accuracy and consistency by optimizing the boundaries of the segmented model. Our proposed system achieved a maximum DICE of about 98 and an average score of about 83.5, while in IoU we achieved a maximum score of about 92 and an average score of about 80. Our proposed method outperformed the work of [18], achieving a score of 92.91 for Method 1 and 93.15% for all ultrasound images. Furthermore, [20] introduced a close-up image segmentation method that combines genes in various environmental conditions, and the results are 95% higher than nia IoU, while our score is 98% higher than the work demand.

CONCLUSIONS

In this paper, we propose a novel method for retinal image segmentation combining fuzzy C-means (FCM) clustering with genetic algorithm (GA) optimization. Experimental results show that the combination of GA with FCM leads to better segmentation accuracy and consistency, as demonstrated by increasingly stable Dice and Inter-Union Competition (IoU) scores across different images. GA helps improve the performance, allowing accurate and reliable measurement of blood vessels, especially in difficult cases where FCM alone does not work well. However, this method has limitations, such as the inclusion of high costs, which may limit its use; the results depend on the quality of the original FCM integration. Future work will focus on the optimization of GA, including additional features in FCM to improve the initial segmentation, using alternative methods for model refinement, and exploring deep learning to support the optimization of GA. Research should focus on improving the computational performance to meet urgent applications, especially in important fields such as medical imaging, which require fast and accurate segmentation. In addition, creating a continuously changing modification of FCM-GA according to the image content will increase the segmentation performance and make the method more powerful and successful in many aspects.

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